PROIECTAREA CU MICROPROCESOARE

C3 2025-2026 Sem. 2

CURS 1 Sapt 1 25 februarie 2026 16:00-19:00

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Regulament disciplina

Nota finala este formata din activitatile:

- Laborator 20%
- Lucrare control 10% **6 / 13 mai 2026**
- Prezentare si sustinere proiect 20%
- Examen 50%
- Bonus prezenta – 10%

Conditii pentru promovare:

- Nota de la laborator trebuie sa fie minim 5 (prezenta obligatorie la toate sedintele de laborator);
- Nota proiect minim 5 (conditie obligatorie pentru intrare in examen);
- Nota de la examen trebuie sa fie minim 5;
- Note de minim 5 la Lucrarea de control.

In cazul reluarii disciplinei intr-un alt an universitar, activitatile nepromovate trebuie parcurse din nou.

Continut disciplina

Familia de Intel 8051 (hw & sw)
Circuite programabile I/O de tip port paralel
Circuite programabile I/O de tip timer
Circuite programabile I/O de tip USART
Circuite programabile I/O pentru controlul intreruperilor
Alte circuite specializate I/O programabile
Alte familii de MCU
Realizarea microsistemelor cu MCU (hw & sw)
Wereable
Harvesting

Bibliografie

Kenneth AYALA, *The 8051 Microcontroller: Architecture, Programming and Applications*, West Publishing Company, 1991

Applied Logic Engineering, *8051 Interfacing and Applications*, 1991

Jack GANSSLE, *The Art of designing Embedded Systems, 2nd ed.*, Newnes, Elsevier, 2008

Ken ARNOLD, *Embedded Controller Hardware Design*, LLH Technology Publishing, 2001

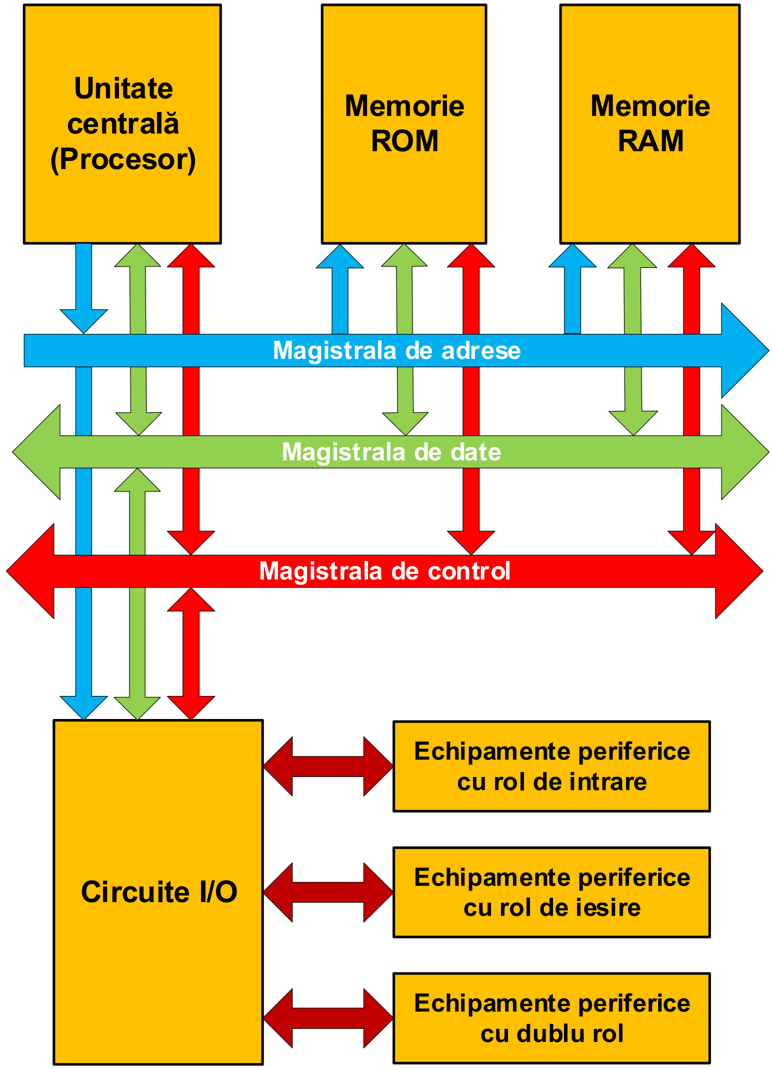
MOSTEK, *Z80 Processor* - *Technical Manual*

Ramesh Gaonkar, *Z-80 Microprocessor Architecture, Interfacing, Programming and Design*,

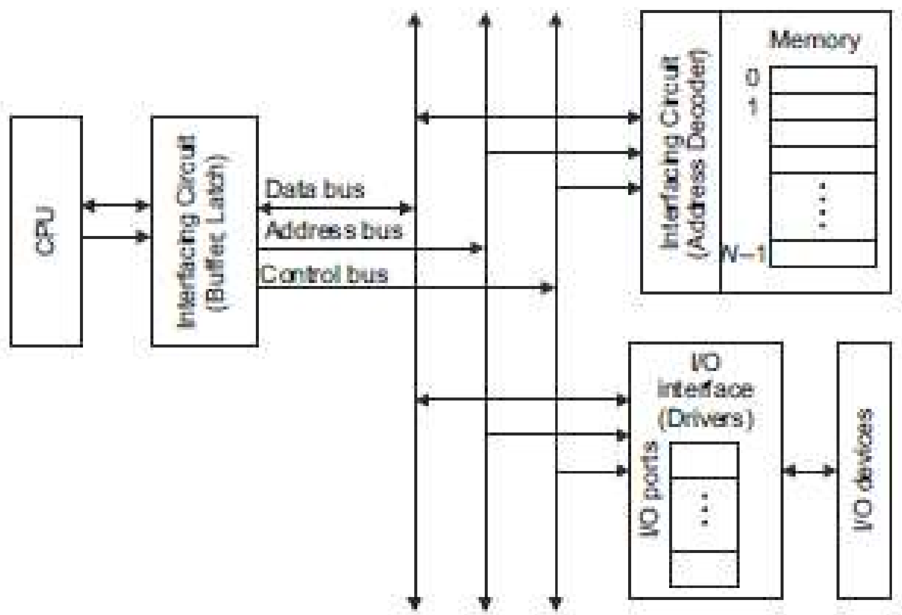
Prentice Hall, 3 ed., 2000

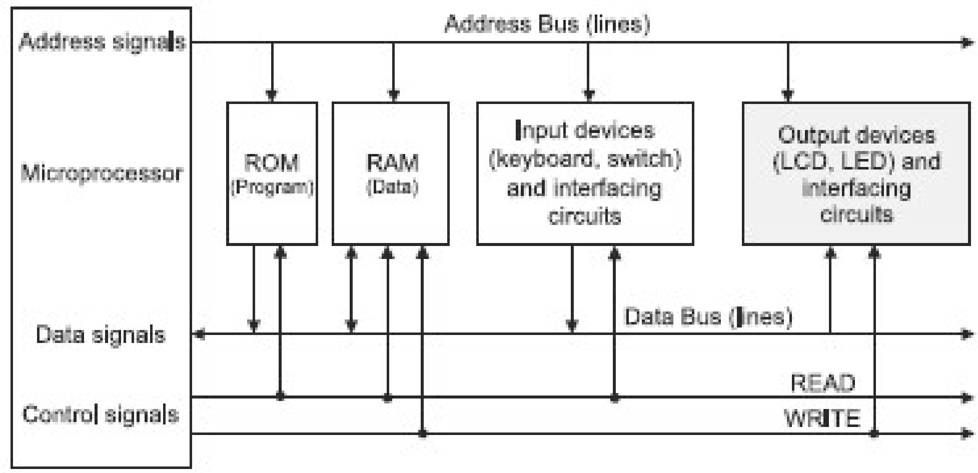
Documentatii de firme (Intel, Zilog)

Structura unui sistem de calcul (arhitectura Von Neumann)



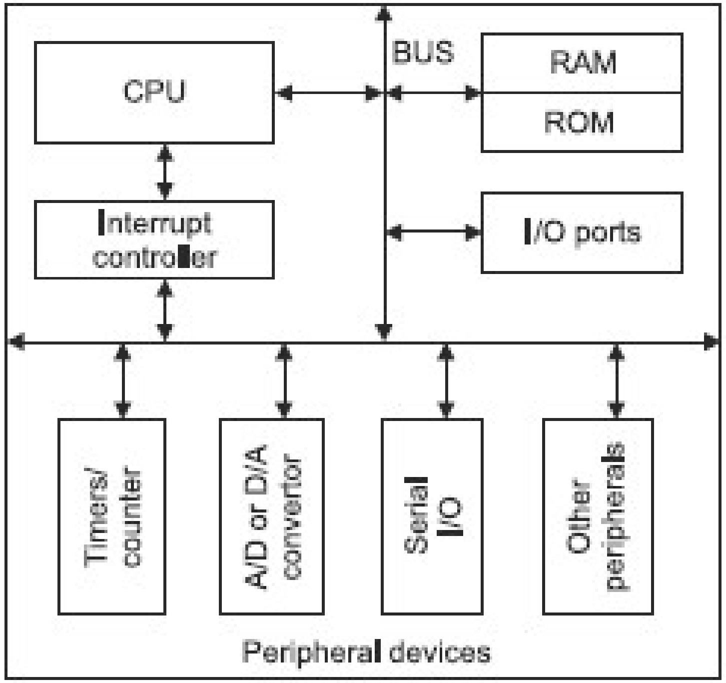
Structura unui sistem de calcul (arhitectura Von Neumann)





Structura unui MCU

Un microcontroler (MCU – MicroController Unit) este un circuit integrat care conține toate componentele digitale necesare realizării si functionarii unui microsistem de calcul: procesor (MPU – MicroProcessor Unit), memorie ROM, memorie RAM, circuite I/O (porturi paralele, canale timer, logică de întreruperi, canale de comunicație serială, convertoare AD și DA, etc).

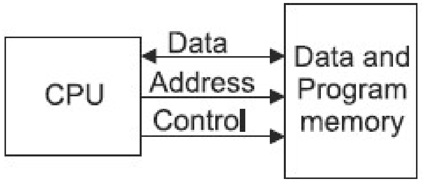


Clasificare MCU

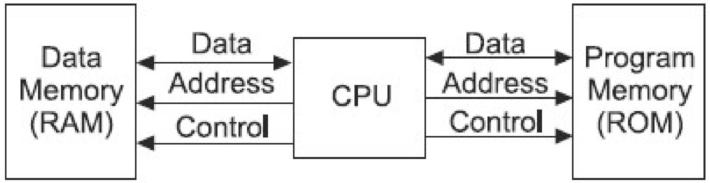
Criterii de clasificare

1). Dupa numarul de biti

2). Dupa arhitectura



Von Neuman (Princeton)

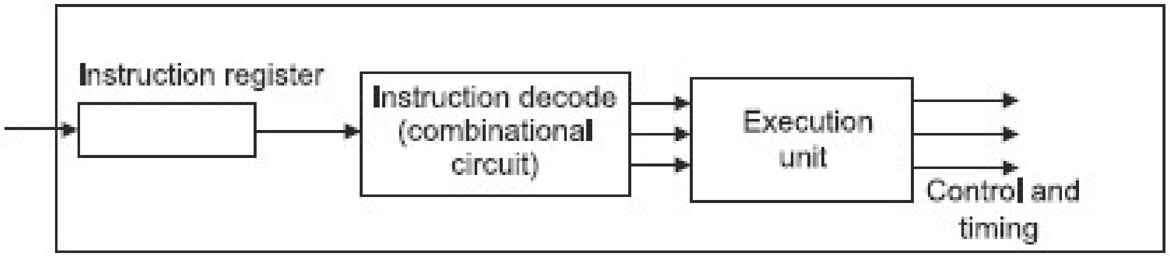


Harvard

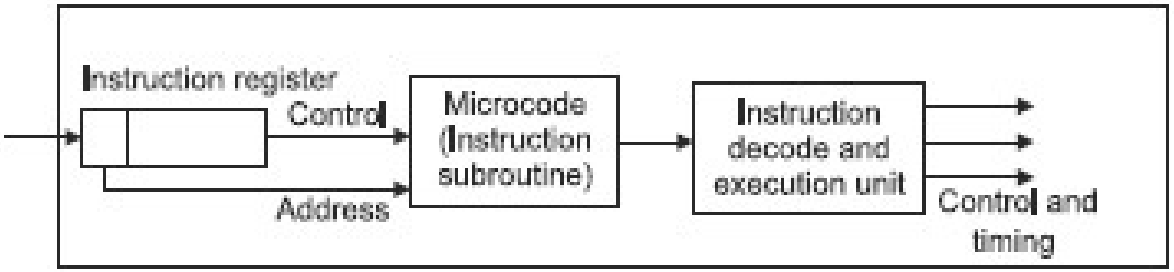
3). Dupa structura setului de instructiuni (RISC, CISC)

Clasificare MCU

4) Dupa modul de executie a instructiunilor



MPU/MCU in Logica cablata (Hardwired logic)



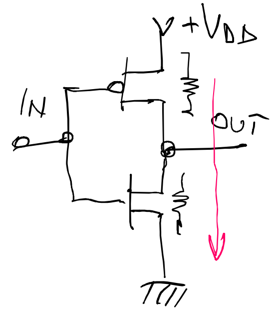
MPU/MCU in Logica microprogramata (Microcoded logic)

FAMILII DE MCU

INTEL | MICROCHIP | ATMEL | MOTORALA

Parameters	MCS 51 (8051)	PIC	AVR	HCS11/12
	89C51	**PIC18F242**	**ATmega32**	**MC68HC912B32**
	89C52	**PIC18F452**	**ATmega64**	**M68HC11K**
	89C54	**dsPIC30F (16 bit)**		
ROM (Flash)	4K	16K	32K	32K
	8K	32K	64K	20K
	16K	Up to 144K		
RAM	128 bytes	768 bytes	2K	1K
	256 bytes	1536 bytes	4K	640 bytes
	256 bytes	8K		
EEPROM	NO	256 bytes	2K	768 bytes
		256 bytes	2K	768 bytes
		4K		
Timers	2(16-bit)	4	3	8
	3(16-bit)	4	4	8
	3(16-bit)	5		
I/O pins	32	34	32	63
		34	53	62
		Up to 54		
Speed up to	33 MHz	40 MHz	16 MHz	25 MHz
		40 MHz	16 MHz	16 MHz
		120 MHz		
Interrupts	5	17	21	24
	6	18	35	22
	6	32		
Power down mode	YES	YES	YES	YES
Watchdog timer	NO	YES	YES	YES
Voltage range	5 V	2 to 5.5 V	4.5 to 5.5 V	6.5 V
				3.0 V to 5.5 V
A to D convertor	NO	10 bit/12 bit	8 CH, 10-bit	8 CH, 10-bit
			8 CH, 10-bit	8 CH, 8-bit
Others	UART	SPI, I2C, PWM In Circuit Debug CAN(dsPIC30F)	JTAG interface, PWM	PWM SPI SCI

CMOS



MCU INTEL 8051

Feature	8031	8051	8751	8032	8052	8752
Program memory	None ROMless	4K ROM	4K EPROM	None ROMless	8K ROM	8K EPROM
Data memory	128 Bytes	128 Bytes	128 RAM	256 Bytes	256 Bytes	256 Bytes
Timers/counters (16-bit)	2	2	2	3	3	3
I/O pins	32	32	32	32	32	32
Serial port	1	1	1	1	1	1
Interrupt sources (Reset not included)	5	5	5	6	6	6

1. 8-bit CPU with Boolean processing capabilities
2. 4K bytes on-chip *program memory
3. 128 bytes on-chip data memory
4. 64 Kbytes each program and external data address space
5. 32 bidirectional I/O lines organized as four 8-bit I/O ports
6. Serial port–Full duplex UART
7. Two 16-bit timers/counters
8. Two-level prioritized interrupt structure
9. Direct byte and bit addressability
10. Four register banks
11. Binary or decimal arithmetic support
12. Hardware multiply and divide operations
13. 12 clock cycles per machine cycle

OTP
one time programming
LOCK BITS

UV

MCU INTEL 8051